EXPERIMENT 1 – SETTING UP PYTHON ENVIRONMENT AND LIBRARIES – JUPYTER NOTEBOOK

Aim

To install Python and Jupyter Notebook, configure the development environment, understand the Jupyter Notebook interface, execute Python programs, create and format Markdown cells, and utilize the basic features of Jupyter Notebook for Data Analytics applications.

Procedure

1. Open Jupyter Notebook and import the Pandas library.
2. Create or place the `student.csv` file in the current working directory.
3. Use the `pd.read_csv()` function to load the CSV file into a Pandas DataFrame.
4. Store the imported data in a variable (e.g., `df`).
5. Display the DataFrame using the `print()` function.
6. Verify that the data has been imported correctly by checking the displayed records.
7. Analyze the imported dataset for further processing if required.

Program

Program 1: Simple Python Program

```
print("Welcome to Data Analytics Lab")
```

Program 2: Arithmetic Operations

```
a = 20
b = 10

print("Addition =", a + b)
print("Subtraction =", a - b)
print("Multiplication =", a * b)
print("Division =", a / b)
```

Program 3: Display Student Details

```
name = "John"
dept = "AI&DS"
cgpa = 8.9

print("Name:", name)
print("Department:", dept)
print("CGPA:", cgpa)
```

Program 4: Creating Markdown Cell

Create a **Markdown** cell (not a Code cell) and type:

```
# Data Analytics Lab
## Experiment 1
```

```
[1]: print("Welcome to Data Analytics Lab")

     Welcome to Data Analytics Lab

[3]: a = 20
     b = 10

     print("Addition =",a+b)
     print("Subtraction =",a-b)
     print("Multiplication =",a*b)
     print("Division =",a/b)

     Addition = 30
     Subtraction = 10
     Multiplication = 200
     Division = 2.0

[4]: name = "John"
     dept = "AI&DS"
     cgpa = 8.9

     print("Name:",name)
     print("Department:",dept)
     print("CGPA:",cgpa)

     Name: John
     Department: AI&DS
     CGPA: 8.9

[5]: # Data Analytics Lab
     ## Experiment 1

     ### Jupyter Notebook
```

Program 5: Using NumPy Library

```
import numpy as np

a = np.array([10, 20, 30, 40, 50])

print(a)
print("Mean =", np.mean(a))
print("Sum =", np.sum(a))
```

Program 6: Using Pandas Library

```
import pandas as pd

data = {
    'Name': ['John', 'David', 'Alex'],
    'Marks': [85, 90, 95]
}

df = pd.DataFrame(data)

print(df)
```

Program 7: Data Visualization

```
import matplotlib.pyplot as plt

x = [1, 2, 3, 4, 5]
y = [10, 20, 15, 25, 30]

plt.plot(x, y)
plt.title("Sample Line Graph")
plt.xlabel("X Axis")
plt.ylabel("Y Axis")
plt.show()
```

```
    Name  Marks
0   John     85
1  David     90
2   Alex     95
```

[11]:
```python
import matplotlib.pyplot as plt
x=[1,2,3,4,5]
y=[10,20,15,25,30]

plt.plot(x,y)
plt.title("Sample Line Graph")
plt.xlabel("X Axis")
plt.ylabel("Y Axis")
plt.show()
```